

Dheeran Shankar

Mississauga, ON | 647-562-6975 | dheeranshankar@gmail.com | github.com/dheeranshankar-del

EDUCATION

York University – Lassonde School of Engineering

Bachelor of Electrical Engineering

Toronto, ON

Sept. 2024 – Apr. 2028

EXPERIENCE

Arbalest Rocketry Club – York University

DAQ & Telemetry Member

Toronto, ON

May 2026 – Present

- Engineered a real-time rocket-to-ground telemetry system using Arduino, Raspberry Pi, UDP networking, Python, and an IMU to enable reliable, low-latency transmission of critical flight data
- Designed and implemented a real-time ground station GUI displaying live telemetry, 3D rocket orientation, attitude data, performance charts, and system health indicators for reliable flight monitoring
- Optimized telemetry transmission using binary packet communication, reducing end-to-end latency by 35%
- Researched, selected, and integrated MCUs, sensors, connectors, and electronic components based on datasheets

Indus Capital Real Estate & Mortgages

Office Administrator

Mississauga, ON

Dec. 2020 – Apr. 2021

- Communicated with 200+ clients through phone, email, and in-office interactions while coordinating appointments, scheduling meetings, and maintaining efficient daily office operations
- Assisted in coordinating and preparing documentation for property closings in residential real estate transactions
- Utilized Microsoft Excel and Word to track client information and organize financial records to support operations
- Coordinated appointments, follow-ups, and communication between clients, agents, and office staff to ensure smooth workflow and timely completion of tasks

PROJECTS

Electrical Fault Detection | *Python, Scikit-learn, XGBoost*

May 2026 – June 2026

- Developed an ML pipeline to detect electrical faults in 3-phase AC systems using voltage and current signal data
- Engineered domain-specific features including impedance, real/reactive power, and voltage/current magnitude to capture electrical behavior, increasing model accuracy from 88.7% to 95.9%
- Evaluated model generalization using cross-validation to assess performance on unseen data and detect overfitting

Autonomous Rover with Sensor Fusion | *C/C++, Python, PlatformIO, ESP32-C3*

Jan. 2026 – Feb. 2026

- Designed and built an autonomous ESP32-based rover capable of navigating unknown environments, avoiding obstacles, and generating a real-time environmental map using sensor fusion
- Integrated an IMU, ultrasonic sensor, temperature sensor, camera module, and dual motor controllers on an ESP32
- Developed low-level C/C++ control logic for PWM motor actuation and real-time sensor interfacing

STM32 Development Board Design | *KiCad, STM32CubeIDE*

Aug. 2025 – Sept. 2025

- Designed a custom 2-layer STM32F103 board for rapid prototyping of embedded hardware and firmware
- Integrated USB Micro-B connectivity, SWD debugging interface, external 16 MHz crystal oscillator, and GPIO expansion headers for embedded development
- Designed a regulated 3.3 V power subsystem with decoupling capacitors, ferrite bead filtering, and power integrity considerations following STM32 hardware design guidelines

Computer Vision-Based Lane Assist for ADAS | *Python, OpenCV, NumPy*

Aug. 2025 – Sept. 2025

- Developed a prototype ADAS lane detection system using OpenCV, Canny edge detection, region-of-interest masking, and Hough Transform line detection
- Processed driving footage at 21 FPS using grayscale conversion, Gaussian filtering, lane extraction, and parameter tuning across varying highway conditions
- Optimized ROI geometry and Hough parameters to improve lane detection and visualization performance

TECHNICAL SKILLS

Programming: C, MATLAB, Python, C++

Tools: LTspice, LabVIEW, VS Code, Git, PlatformIO, KiCad, Linux, oscilloscopes, breadboarding

Hardware: Raspberry Pi, ESP32-C3, Arduino Uno, STM32F103, PCB design, PCB soldering & assembly